

Federated DDoS Detection with Clustered Quantization-Aware Training Models for IoRT

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Abstract—IoRT systems, which operate with microcontroller-class processors and limited memory ($\leq 2MB$), require real-time DDoS detection. However, existing federated learning approaches cannot simultaneously achieve the necessary model compression, DP, and communication efficiency for resource-constrained deployments. This paper presents a novel federated learning framework that resolves these conflicting requirements through an integrated QAT approach with DP-based clustering.

Our approach achieves 4.0x model compression (0.52MB to 0.13MB) with only 0.79% accuracy degradation compared to uncompressed federated baselines, while maintaining F1-scores (0.998). Evaluation against methods such as FedProx, DeepShield, and AUWPAE on CICIoT2023 demonstrates the advantage of our approach: 433x faster inference than centralized ensemble methods (0.15ms vs. 65ms) with a 75% reduction in federated communication overhead. The DP clustering successfully handles non-IID data heterogeneity while providing formal privacy guarantees unavailable in centralized approaches.

Index Terms—Internet of Robotic Things (IoRT), Distributed Denial of Service (DDoS), Federated Learning, Quantization-Aware Training (QAT), DP, Non-IID Data, Model Compression

I. INTRODUCTION

Robotic devices are increasingly interconnected in IoRT systems to enable collaborative intelligence and coordinated operations [1]. However, this connectivity exposes IoRT networks to cyber threats such as DDoS attacks [2]. Unlike single-source DoS attacks, which can be mitigated through rate limiting or IP blacklisting, Distributed Denial-of-Service (DDoS) attacks use distributed botnets to generate overwhelming traffic from multiple sources. While internet-scale attacks can reach terabits per second [3], even localized attacks measured in gigabits can be sufficient to saturate the constrained network links common in industrial or medical IoRT deployments, disrupting time-sensitive operations.

IoRT devices face unique resource constraints and real-time requirements that differentiate them from general IoT environments. Mobile robots typically employ microcontroller-

class processors with limited RAM and storage, making them susceptible to traffic floods [4]. With safety-critical control loops requiring response times of 50ms or less, network saturation can cause unsafe actuation before a remote gateway or cloud-based mitigation responds. This necessitates local, on-device detection.

Deploying effective federated models on-device faces a challenge: balancing model compression with privacy preservation. Standard Differential Privacy (DP) introduces noise into model gradients, which disrupts compression techniques like Quantization-Aware Training (QAT). We address this conflict with a novel federated architecture that separates noise injection from model weights. By integrating DP at the aggregation stage through privacy-preserving client clustering, we protect client privacy without undermining the precision needed for effective INT8 quantization. This approach, along with FedProx regularization for stable training, achieves 4.0x model compression while ensuring strong privacy and high detection performance.

Our contributions are:

- A novel federated learning framework that resolves quantization-privacy conflicts by integrating QAT with DP-based clustering.
- A compact 0.13MB detector that classifies traffic in less than 0.2ms with 99.8% F1-score.
- Comprehensive evaluation demonstrating high detection accuracy and efficiency gains.

The remainder of this paper is organized as follows: Section II surveys related works; Section III defines the threat model; Section IV details the methodology; Section V outlines the experimental setup; Section VI discusses results; Section VII addresses limitations, and section VIII concludes.

II. RELATED WORKS

Our work addresses a fundamental integration challenge at the intersection of DDoS detection in IoRT, resource-

efficient model compression, and privacy-preserving federated learning. Existing research typically addresses these challenges in isolation, leading to solutions that are unsuitable for the simultaneous constraints of real-world IoRT deployments.

A. DDoS Detection in IoT and IoRT Systems

Recent efforts in IoT intrusion detection have primarily followed a centralized training paradigm. For instance, DEEPShield [5] uses a deep ensemble with model pruning for edge IoT networks, and Beshah *et al.* [6] propose an adaptive ensemble to handle concept drift. While effective in their respective contexts, these centralized approaches require raw data to be sent to a central server, which introduces significant privacy risks and communication overhead unsuitable for distributed IoRT systems.

Recognizing these limitations, researchers have explored federated learning. Nkoom *et al.* [2] demonstrated that client clustering can effectively manage non-IID data distributions in federated DDoS detection. Other federated approaches [7] have also achieved high accuracy on IoT datasets. However, these works largely overlook the severe resource constraints of IoRT devices, often proposing models that are too large or computationally expensive for on-device inference. They establish the need for federated, non-IID-aware learning but stop short of making it feasible for microcontroller-class hardware.

B. Resource-Efficient Model Compression

To deploy models on resource-constrained devices, compression techniques like pruning and quantization are essential. Pruning as used in DEEPShield [5], reduces model size by removing less important connections. However, it often yields limited compression ratios because the remaining parameters are kept at full precision.

Quantization, which converts model weights and activations to low-precision integers, offers far greater compression and latency improvements. QAT is effective, as it simulates low-precision effects during training to minimize accuracy loss [8]. While QAT is a powerful tool for optimizing standalone models, a critical research gap remains in integrating it directly into the federated learning process for IoRT security, where communication efficiency, privacy, and model performance must be co-designed.

C. The Privacy-Quantization Conflict in Federated Learning

Integrating privacy mechanisms like DP with quantization in federated settings presents a fundamental conflict. The standard method for achieving DP in federated learning involves injecting statistical noise directly into model gradients before they are sent to the server. This process fundamentally disrupts the precise statistical distributions (e.g., mean and variance of weights) that QAT relies on for calibrating quantization parameters, leading to severe model degradation.

Existing works treat model compression, privacy, and non-IID data handling as separate optimization problems. This leads to suboptimal solutions where privacy noise destroys

quantization accuracy, or federated algorithms cannot be compressed for on-device deployment. Our work addresses this fundamental integration challenge directly.

III. THREAT MODEL

We consider an IoRT deployment where robots have microcontroller-class processors, up to 512kB of RAM, and limited megabytes of memory, communicating over bandwidth-constrained 802.11 or LPWAN links. These robots function in safety-critical real-time environments, where delays or disruptions can be dangerous. Accordingly, any on-device defensive mechanisms must: (i) use less than 2 MB of memory and (ii) complete inference within 50ms.

We consider three adversary types based on threat source and trust level between learning participants.

- **External attackers** lack valid device certificates and can only engage at the network level. Their goal is to disrupt operations by generating volumetric attacks (e.g., UDP floods) and protocol attacks (e.g., SYN floods). While they cannot directly alter device software, even minor network disruptions can lead to unsafe actuator behavior or mission failure.
- **Honest-but-Curious Insider Participants** are authenticated devices following protocols, but may seek to extract sensitive information from updates. Their goal is to compromise the privacy of other participants by analyzing the aggregated model updates. They may attempt membership inference attacks to determine if a specific device participated in a training round or statistical disclosure attacks to learn sensitive features about another robot's network traffic.
- **Model-poisoning (Byzantine) participants** are compromised, authenticated devices that intentionally submit malicious weight updates to corrupt the global model's performance. We acknowledge that a sophisticated attacker who gains control of a device would possess a valid identity. However, this work's primary focus is on the quantization-privacy conflict under the honest-but-curious model. We therefore assume Byzantine participants constitute a small fraction (less than 10%) of the fleet.

The proposed system is therefore designed to provide real-time, on-device detection of external DDoS attacks while ensuring the privacy of participants' data against honest-but-curious peers within a resource-constrained federated environment.

IV. PROPOSED SOLUTION

Our framework integrates clustered federated learning with QAT through five key stages: neural network design, data preprocessing, differentially private client clustering, and quantized federated training.

A. Convolutional Neural Network Architecture

To balance performance and on-device constraints, we employ a 1D CNN. The model processes a 30-feature input vector through two 64-channel convolutional layers (kernel=3,

Algorithm 1 Non-IID Data Partitioning

Require: Dataset D with labels, #clients N , Dirichlet parameter α

Ensure: Client indices $\{S_i\}_{i=1}^N$

- 1: **Compute class-wise indices:**
- 2: **for** each class k in labels **do**
- 3: $I_k \leftarrow \{j : D[j].\text{label} = k\}$
- 4: **end for**
- 5: **Sample Dirichlet proportions:**
- 6: **for** each class k **do**
- 7: Draw $\pi_k \sim \text{Dirichlet}(\alpha, \dots, \alpha)$ ▷ length N vector
- 8: **end for**
- 9: **Assign samples to clients:**
- 10: **for** each class k **do**
- 11: Shuffle I_k
- 12: Partition I_k into N chunks of sizes $\lfloor \pi_{k,i} |I_k| \rfloor$
- 13: **for** client $i = 1, \dots, N$ **do**
- 14: Append chunk $_{k,i}$ to S_i
- 15: **end for**
- 16: **end for**
- 17: **return** $\{S_i\}$ ▷ Each S_i is non-IID across classes

padding=1), each followed by ReLU activation and max-pooling (size=2). A dropout layer (0.25) is applied before features are flattened (960 units) and passed to a 128-unit dense layer with ReLU and another dropout (0.5). The final layer is a 7-class softmax classifier. This architecture, totaling 136,519 parameters (0.52MB pre-quantization), was specifically designed for compatibility with QAT.

B. Data Preprocessing and Non-IID Distribution

Network traffic features first undergo standard normalization and standardization. To simulate the data heterogeneity(non-IID) of real-world IoRT deployments, we partition the dataset among clients using a Dirichlet distribution($\alpha=0.5$). As detailed in Algorithm 1, this method samples class proportions for each client from the Dirichlet distribution, ensuring each client receives a statistically varied and skewed subset of the data that reflects realistic traffic patterns.

C. Differentially-Private Clustering for QAT

A core innovation of our work is resolving the conflict between DP and QAT. Standard DP injects noise directly into gradients, which corrupts the statistical distributions QAT relies on for calibration. We solve this by decoupling noise injection from the gradient update path.

Instead of adding noise during training, we apply DP in a pre-training clustering phase based on the method from [9]. Each client trains a preliminary model on local data for a few epochs, computes lightweight statistical summaries (such as quantiles), and injects calibrated noise to ensure DP. The server receives these noisy summaries to cluster clients with similar data distributions.

We set the privacy budget for this one-time clustering step to $\epsilon = 1$, balancing privacy and cluster utility. This standard in

applied DP provides robust protection against privacy attacks while allowing for meaningful client clusters. A lower ϵ would enhance privacy but may degrade cluster quality and model accuracy, while a higher ϵ would do the opposite. Our results, showing high model accuracy, confirm that this budget is effective for the task.

This 'cluster-then-train' approach ensures that the noise required for privacy is only applied to the clustering metadata, not to the model weights themselves. The subsequent QAT-based federated training can then proceed within each cluster using noise-free gradients, preserving the numerical precision required for effective quantization.

D. Quantized Federated Training with FedProx

Once clients are clustered, we initiate the main federated training process, where a separate global model is trained for each cluster using Algorithm 2. This stage integrates QAT for efficiency and FedProx for stability.

- 1) **Quantization-Aware Training :** We apply QAT [8] to enable resource-efficient on-device detection. QAT simulates the effect of quantization during training by inserting simulated quantization operations(float $\rightarrow$ int $\rightarrow$ float) into the model. This allows the model to learn to be robust to the precision loss. We quantize both weights and activations to 8-bit signed integers ($q_{min} = -128$ and $q_{max} = 127$) using per-tensor symmetric quantization (zero-point $z = 0$), as defined by :

$$q = \text{clamp} \left(\text{round} \left(\frac{r}{s} \right) + z, q_{min}, q_{max} \right) \quad (1)$$

We use PyTorch's QAT framework [10], applying quantization to all layers before training. Observer modules track tensor statistics during training to determine the optimal scaling factor s for each layer, preserving the dynamic range of the values within the 8-bit range.

- 2) **FedProx Integration for QAT Stability:** We observed that QAT alone can cause training instability, leading to significant accuracy degradation (9-15%) compared to the full-precision model. To counteract this, we integrate the FedProx [11] regularization term. The local client loss function L_i is modified to include a proximal term that penalizes large deviations from the cluster's global model w^t :

$$L_i(w) = F_i(w) + \frac{\mu}{2} \|w - w^t\|^2$$

This term stabilizes the local QAT process by preventing excessive client-side drift caused by quantization noise. Our experiments show this integration restores performance to within 2% of the full-precision baseline.

The overall training procedure, outlined in Algorithm 2, initializes a separate server for each cluster. In each round, clients perform local training using a QAT-enabled model and the FedProx objective function. Updates are then aggregated only within their respective clusters, maintaining data separation while allowing specialized models to develop for each data distribution.

Algorithm 2 Clustered FL with FedProx & QAT

Require: Client splits $\{S_i\}$, cluster IDs $C[i] \in \{1..K\}$, model factory $\mathcal{F}(\text{QAT_flag})$, local epochs E , rounds T , proximal μ

Ensure: trained cluster models and metrics

```

1: Initialize servers & clients per cluster:
2: for each cluster  $c = 1..K$  do
3:    $(m_c, \cdot, \cdot) \leftarrow \mathcal{F}(\text{QAT\_flag})$ 
4:    $S_c \leftarrow \text{Server}(m_c)$ 
5:    $L_c \leftarrow \{\text{Client}(m_c, S_i) \mid C[i] = c\}$ 
6: end for
7: for round  $r = 1$  to  $T$  do
8:   for each cluster  $c$  do
9:      $\theta^r \leftarrow S_c.\text{get\_params}()$ 
10:    for each client  $i \in L_c$  do
11:       $i.\text{set\_params}(\theta^r)$ 
12:       $(\Delta_i, n_i) \leftarrow i.\text{train}(E, \theta^r, \mu) \triangleright \text{FedProx adds } (\mu/2)\|\theta_i - \theta^r\|^2$ 
13:    end for
14:     $S_c.\text{aggregate}(\{\Delta_i\}, \{n_i\})$ 
15:    Evaluate  $S_c$  on validation set
16:  end for
17: end for
18: for each cluster  $c$  do
19:   Evaluate  $S_c$  on test set; record metrics (loss, acc, F1)
20:   Measure inference time & model size
21: end for
22: return All per-cluster metrics

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V. EXPERIMENTS

A. Experimental Setup

The experiments were conducted in a simulated IoRT environment with 20 clients, a scale representative of a single industrial floor or robotic fleet. Clients were distributed across 3 to 6 clusters based on data distribution similarity. The environment utilized PyTorch 1.12.0 on an Intel Core i7 system with 64 GB of RAM, using only CPU processing to reflect on-device capabilities. Key hyperparameters, including a learning rate of 0.00031, FedProx μ of 0.1, and 2 local epochs, were determined through grid search on the validation set. In the federated setup, we employed a Dirichlet distribution $\alpha = 0.5$ to create a realistic non-IID data scenario.

B. Dataset and Preprocessing

We evaluated our model on the CICIoT2023 dataset [12], which features 33 attack types from 105 heterogeneous IoT devices. After merging the 63 CSV files, the data was cleaned by removing infinite values and using forward-fill for missing data. We then performed feature engineering, removing nine highly correlated features and using Random Forest analysis to select the 30 most informative features.

While the dataset is comprehensive, our evaluation is intentionally focused on DDoS attacks. This design choice is directly motivated by our threat model III, where high-volume,

low-latency DDoS attacks pose the most immediate and critical threat to the safety-critical control loops of the IoRT systems. Other attack classes represent different threat vectors that fall outside the primary scope of this real-time network saturation defense framework.

C. Evaluation Metrics

We evaluated our approach using detection performance (accuracy, F1-score), resource utilization(model size and inference time). Additionally, we assessed model convergence through training and validation loss curves across federated rounds.

D. Baselines and Comparative Methods

We compare our full model, DP_Clustering + QAT+ FedProx, against several key baselines to isolate the impact of each contribution:

- 1) DP_Clustering + FedProx: An ablation baseline using the full-precision model to measure the raw impact of QAT.
- 2) FedProx_Baseline: A standard FL baseline without clustering or QAT, used to quantify the benefits of our clustering-based approach.
- 3) Centralized State-of-the-Art: We also benchmark against three centralized methods: DeepShield (unpruned), the DeepShield_Pruned variant [5], and AUMPAE [6], a drift-adaptive ensemble method.

This comparison highlights the trade-offs in accuracy, efficiency, and privacy, demonstrating the necessity of a purpose-built federated solution.

VI. RESULTS

A. Classification Performance

As shown in Table I, all federated models demonstrated excellent detection performance, achieving competitive accuracy while providing the vital privacy guarantees required for IoRT deployments.

Our full model, DP_Clustering+QAT+FedProx, achieved 98.97% accuracy, a minimal and acceptable degradation of only 0.79% compared to its unquantized counterpart, DP_Clustering+FedProx (99.76). Critically, both models maintained an identical F1-score of 0.998, demonstrating that the 4.0x compression did not impact the model's ability to handle class imbalance. In contrast, the standard FedProx_Baseline achieved a similar accuracy (99.76%) but a noticeably lower F1-score(0.992). This suggests that our DP-clustering approach, by grouping similar clients, provides a significant benefit in handling the non-IID data distributions, leading to a more robust model.

The centralized methods AUWPAE and DeepShield achieved slightly higher accuracies at 99.8% and 99.58%, respectively. However, they are fundamentally unsuitable for this IoRT problem as they lack the privacy guarantees and require centralizing sensitive data. Furthermore, AUWPAE highlights a critical performance trade-off. Despite its tiny 0.02MB footprint, its inference time is 65.1ms. This result is

TABLE I: Comparison with Federated and Centralized Baselines

Approach	Accuracy	F1-score	Size (MB)	Inference (ms)	Training (min)
DP_Clustering+FedProx [9]	0.9976	0.998	0.52	0.13	248.6
FedProx Baseline [11]	0.9976	0.992	0.52	0.53	270.0
DP_Clustering+QAT+FedProx	0.9897	0.998	0.13	0.15	379.87
DeepShield [5]	0.9958	99.52	0.31	0.13	156
DeepShield_Pruned [5]	0.9957	0.9961	0.21	0.156	241.53
AUWPAE [6]	0.998	0.998	0.02	65.138	124.51

due to its ensemble architecture, which requires a computationally expensive averaging process for every inference, making it 433x slower than our approach and non-viable for our less than 50ms real-time requirement.

Convergence analysis in Figure 1 indicated that the FedProx_Baseline reached 95% validation accuracy in 3 rounds, while our quantized method needed 7 rounds. This slower convergence is an expected and acceptable one-time training trade-off, as it allows the QAT process to stabilize, and is offset by the permanent efficiency gains.

B. Efficiency Analysis

Our DP_Clustering+QAT+FedProx approach achieved the primary goal of resource efficiency. The model size was reduced by 75% from 0.52MB to just 0.13MB. This 4.0x compression aligns with the theoretical expectations of INT8 QAT and directly translates to a 75% reduction in communication overhead per round.

Inference performance remained competitive after quantization. The unquantized DP_Clustering+FedProx model achieved an inference speed of 0.13ms, while our final quantized model took 0.15ms. Both are well within our 50ms real-time safety constraint and dramatically faster than the non-viable 65.1ms of the AUWPAE baseline.

Training time analysis shows that our QAT implementation took 379.87 minutes, compared to 248.6 minutes for its unquantized counterpart. This added time reflects the computational overhead is a one-time training cost that is clearly offset by the long-term, permanent advantages of a 4.0x smaller model size, 75% less communication overhead, and extremely fast on-device inference.

C. Summary of Key Findings

Our findings reveal that our framework is the first to integrate QAT with privacy-preserving federated learning, achieving 4.0x model compression with only a 0.79% accuracy drop. The final 0.13MB model has an inference time of 0.15ms, meeting our 50ms safety-critical requirement and outperforming the AUWPAE ensemble by 433x. Additionally, the combination of FedProx and DP-clustering allows the QAT model to converge with a 0.998 F1-score on non-IID data while ensuring privacy guarantees not offered by centralized methods.

VII. LIMITATIONS AND FUTURE WORK

Our study has several limitations that define a clear path for future research.

We focused on DDoS attacks within a single dataset, CIIoT2023. Future work will validate our framework's generalizability by testing it on additional datasets and against a wider variety of attack classes.

Additionally, our results are simulation-based. A critical next step is to deploy and benchmark the framework on actual resource-constrained IoRT hardware (example, ARM Cortex-M devices). This would not only validate our latency figures but also allow for a rigorous analysis of energy-per-inference.

We acknowledge QAT's increased training time and did not analyze scalability beyond 20 clients. Future work should investigate the training overhead and clustering performance in much larger federations.

Finally, our analysis used model size reduction as a proxy for efficiency but provided no detailed metrics on the actual data transmitted. A future robust analysis must move beyond this proxy to measure the true communication costs, including overhead from aggregation protocols, to fully validate the system's efficiency for IoRT networks.

VIII. CONCLUSION

This paper addresses the challenge of real-time DDoS detection for resource-constrained IoRT devices. We introduced a novel federated learning framework that successfully resolves the fundamental conflict between QAT, DP, and non-IID data handling. Our approach integrates QAT with a DP-clustering mechanism, delivering a high-performance detector that is both practical and efficient. The key quantitative achievements of this work include a 4.0x model size reduction with a negligible 0.79% accuracy drop, an exceptionally fast 0.15ms inference time, which is 433x faster than complex centralized ensembles, and a framework that maintained a high F1-score (0.998) on non-IID data while providing formal privacy guarantees that centralized state-of-the-art methods lack.

This work provides a foundational and viable architecture for deploying secure, private, and efficient collaborative intelligence in real-world IoRT systems. While future work will broaden the generalizability of this framework, our results establish a new baseline for what is possible in on-device federated learning under simulations, conflicting constraints.

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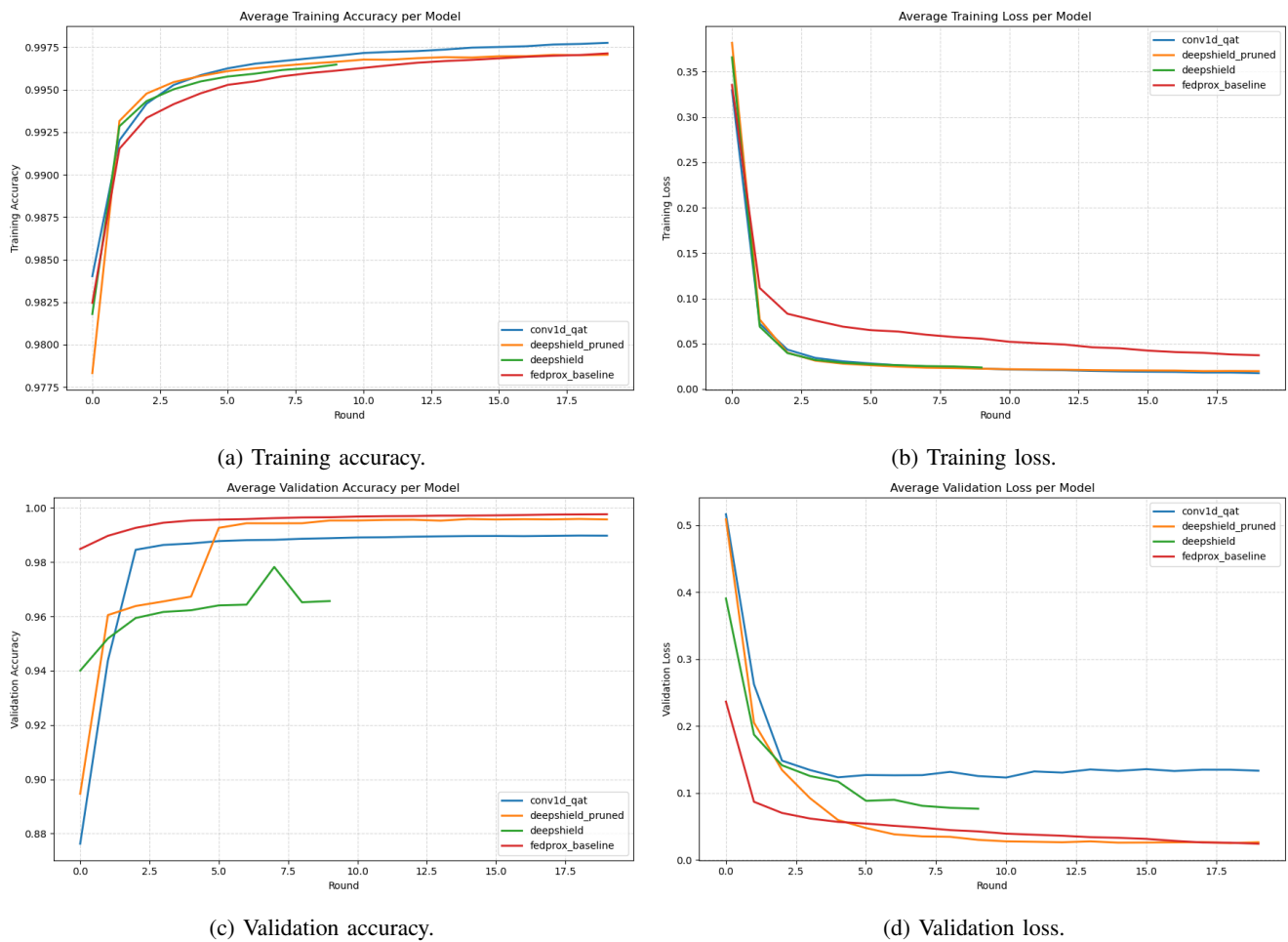


Fig. 1: Federated learning convergence curves for the evaluated approaches.

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